

GenAI Cost Comparison

Summary Tables — Claude API, Azure AI Foundry, Small Local Models

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Companion summary to the full GenAI Provider Cost Comparison & Business Case document.

1. Claude API (Anthropic, direct)

Pricing per 1 million tokens, as published September 2026.

Model	Input \$/1M tok	Output \$/1M tok	Best use in this project
Haiku 4.5	\$1.00	\$5.00	Cheap first-pass: tagging, extraction, chunk summarising
Sonnet 5	\$2.00	\$10.00	Default drafting / synthesis model
Opus 5	\$5.00	\$25.00	Final-pass polishing of key sections

Annual cost at InnoBeweegLab's scale

Three usage scenarios sized to a periodic report-generation workflow (not a high-traffic chat app), compared against the two other leading frontier options.

Model	Light /yr	Medium /yr	Heavy /yr	Notes
Claude Sonnet 5	\$12	\$60	\$380	Anthropic direct
GPT-5.6 Terra	\$12	\$62	\$396	OpenAI direct or Azure Foundry
Mistral Large 3	\$3	\$14	\$87	Cheapest EU-native frontier option

Light = 6 reports/yr (~800K input / 40K output tokens each). Medium = 20 reports/yr (~1.2M input / 60K output tokens each). Heavy = 30 reports/yr plus daily interactive use by a ~5-person research team (150M input / 8M output tokens/yr).

KEY INSIGHT

Pure model inference cost is negligible in every scenario — even "Heavy" usage across an entire 5-person team lands under €400/year for the model itself. Data residency, not price, is the deciding factor for Claude direct: the first-party API has no published EU region, so identifiable resident data should not be sent to it directly (see Section 3).

2. Azure AI Foundry

Microsoft's hosting layer for models from OpenAI, Anthropic, Meta, Mistral and others, plus enterprise plumbing (Entra ID SSO, Azure AI Search for retrieval, private networking). Two costs apply: the per-token model price, and the infrastructure needed to integrate it.

2.1 Model pricing on Foundry

Provider	Model	Input \$/1M	Output \$/1M	Notes
Azure AI Foundry	GPT-5.6 Terra	\$2.00	\$12.00	Same model as direct API, EU region selectable
Azure AI Foundry	GPT-4o mini	\$0.165	\$0.66	Very cheap legacy option, still supported
Azure AI Foundry	Claude Sonnet 4.5	\$3.00	\$15.00	Avoid for personal data — no EU region

2.2 Integration cost: lean path vs. enterprise path

Foundry itself has no platform fee — cost comes from the Azure resources provisioned underneath it. A lean, self-built path suits a 4-month prototype far better than a fully managed enterprise stack.

Component	Lean-path cost	Why
Azure OpenAI model calls (via Foundry)	≈ €1–35/month	Same Light/Medium token pricing modelled in Section 1 — negligible at this volume
App hosting (Azure Functions, Consumption plan)	€0	1,000,000 free executions + 400,000 GB-seconds included monthly — well above what a periodic job needs
Document storage (Blob Storage)	Cents/month	A few GB of transcripts and datasets
Retrieval index (Azure AI Search, Free F0 tier)	€0	Free tier capped at 50MB / 3 indexes — enough for the 4-month build
Identity / SSO (Entra ID)	€0	InnoBeweegLab's existing Microsoft 365 tenant already includes Entra ID Free
Private networking	€0 (skipped)	Not required at prototype scale — API key/IP restriction is sufficient

COST LEVER

Lean-path total: roughly €5–50/month — not the €500–1,000+/month figure typically quoted for Azure AI Foundry, which assumes a fully managed, production-hardened stack (Search Basic tier, private endpoints on every service, higher-tier monitoring). That tier buys SLA guarantees and growth headroom that a capstone prototype does not need yet.

2.3 Data residency by provider / route

Provider / route	EU residency	Notes
Anthropic Claude — direct API	Not guaranteed	No native EU region; EU routing would need AWS Bedrock or Google Vertex AI instead
OpenAI — direct API	Optional, +10%	Data-residency setting at a price premium; otherwise processed outside the EU
Azure OpenAI (native, via Foundry)	Yes	West Europe (NL) or North Europe (IE) — most straightforward EU-compliant path in the Microsoft ecosystem
Claude on Microsoft Foundry	No	GA since mid-2026 but zero EU regions — do not route resident data through this combination
Mistral AI	Yes, by default	EU/French company and infrastructure — no special configuration needed
Self-hosted (local)	Yes, fully	Data never leaves InnoBeweegLab-controlled infrastructure

3. Small Local Models (free self-hosting)

Free tools (Ollama, or LM Studio for a GUI) run open-weight models on whatever hardware is available, at zero licensing cost. What fits depends on the machine — directly relevant to the project's "16GB office laptop" constraint.

Hardware	Model that fits	Notes
Typical laptop — 16–32GB RAM, no dedicated GPU	Llama 3.1 8B, Qwen3 8B, Phi-4 Mini	Usable but slow: roughly 12–25 tokens/second on CPU
Gaming GPU (12–16GB+ VRAM) or Apple Silicon (18GB+ unified memory)	Qwen3 14B, Mistral Small 3.1 24B	Noticeably better quality and much faster generation
70B-class models	Needs 40GB+ VRAM or 64–128GB unified memory	Out of reach for typical student/office hardware — treat as unavailable for this project

For heavier occasional runs, Kaggle gives every account roughly 30 hours/week of free GPU time (a 16GB P100, or dual-T4 with 32GB combined), sessions capped at 9 hours, no credit card required. It fits a periodic batch report job well — no persistent server needed.

DESIGN IMPLICATION

A free local model is not just the cheapest option — it is the strongest option for any pipeline step that touches raw, identifiable resident data. Run locally, that data never crosses the internet, sidestepping the GDPR/residency question in Section 2.3 rather than managing it. Recommended design: run a local model (Ollama, or a Kaggle batch job) for anonymisation, tagging, and structured extraction from raw transcripts — where sensitivity matters most and prose quality matters least — then hand only the already-anonymised output to a frontier hosted model (Claude Sonnet 5, or Azure OpenAI in an EU region) for the final drafting/synthesis pass.

Figures as published by each provider, and third-party pricing/hardware analyses (nOps, CloudZero, PromptQuorum, LeanLM), September 2026. Full source list in the companion GenAI Provider Cost Comparison & Business Case document, Appendix 11.3. LLM API pricing changes frequently — re-check before procurement.